

TRIBHUVAN INTL, Nepal

WMO: 444540

Lat: 27.697N

Lon: 85.359E

Elev: 1338

StdP: 86.25

Time Zone: 5.75 (NPL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 3.1	(d) -1.2	(e) 4.0	(f) 18.3	(g) 0.3	(h) 4.5	(i) 11.7	(j) 5.4	(k) 15.5	(l) 5.1	(m) 15.7	(n) 0.5	(o) 90	(p) 0.279

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 8.7	(c) 30.8	(d) 20.6	(e) 29.9	(f) 20.8	(g) 29.0	(h) 21.3	(i) 24.2	(j) 28.4	(k) 23.7	(l) 27.7	(m) 23.3	(n) 27.1	(o) 3.7	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 23.1	(b) 21.0	(c) 27.2	(d) 22.5	(e) 20.2	(f) 26.2	(g) 22.1	(h) 19.8	(i) 25.6	(j) 81.8	(k) 28.5	(l) 79.3	(m) 28.1	(n) 77.3	(o) 27.4	(p) 28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.2	(b) 5.3	(c) 4.6	(e) 0.1	(f) 32.5	(g) 1.9	(h) 1.1	(i) -1.3	(j) 33.3	(k) -2.4	(l) 34.0	(m) -3.5	(n) 34.6	(o) -4.9	(p) 35.4		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	10.7	13.7	17.0	20.3	22.4	24.2	24.2	24.3	23.4	20.4	16.0	11.9		
	DBStd	5.08	1.52	2.03	2.09	1.87	1.80	1.36	1.07	1.04	1.25	2.13	1.85	1.63		
	HDD10.0	13	8	1	0	0	0	0	0	0	0	0	0	4		
	HDD18.3	699	236	130	50	5	1	0	0	0	0	6	73	199		
	CDD10.0	3318	31	105	218	308	384	426	439	442	401	322	179	63		
	CDD18.3	963	0	1	10	63	126	176	181	184	151	70	2	0		
	CDH23.3	5542	1	14	161	672	926	1051	816	851	658	361	32	0		
(13)	CDH26.7	1148	0	1	16	169	246	252	134	167	112	52	1	0		
(14)	Wind	WSAvg	1.5	1.2	1.5	1.8	1.9	2.0	1.7	1.4	1.3	1.4	1.1	1.1		
(15) Precipitation	PrecAvg	1543	12	22	36	58	140	267	383	359	206	68	4	13		
	PrecMax	2108	30	53	98	184	213	658	518	691	376	203	25	62		
	PrecMin	1083	0	0	0	0	39	87	257	222	57	0	0	0		
	PrecStd	317	11	15	31	43	50	143	85	109	94	57	7	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	25.7	28.8	31.2	31.9	31.9	30.2	30.8	30.2	29.6	25.9	22.1		
		MCWB	13.3	15.5	16.3	16.5	20.8	20.9	23.9	24.4	23.3	22.1	18.0	13.9		
	2%	DB	19.9	23.2	26.8	30.0	30.4	30.1	29.1	29.2	29.1	28.1	24.2	20.9		
		MCWB	11.9	13.7	16.0	16.4	19.5	22.1	23.6	23.7	22.7	20.8	16.8	13.2		
	5%	DB	18.2	22.0	25.1	28.8	29.2	29.1	28.2	28.4	28.1	26.9	23.0	19.5		
		MCWB	10.9	13.1	14.9	16.4	19.6	21.9	23.2	23.5	22.4	20.3	16.0	12.4		
	10%	DB	17.1	20.2	23.9	27.2	28.0	28.1	27.2	27.7	27.1	25.3	21.9	18.0		
		MCWB	10.3	12.4	14.3	16.5	19.6	22.0	22.8	23.2	21.9	19.2	15.4	11.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	17.6	19.7	21.2	24.0	24.3	24.7	25.0	24.3	23.2	19.4	15.1		
		MCDB	19.0	23.8	24.6	28.1	29.4	28.7	28.7	28.9	28.8	28.5	24.3	20.4		
	2%	WB	12.7	15.5	18.0	20.1	22.4	23.5	24.1	24.2	23.4	21.9	17.6	13.8		
		MCDB	18.1	21.0	22.4	25.9	27.8	27.9	28.0	28.3	27.7	26.6	23.1	19.2		
	5%	WB	11.7	14.3	16.7	19.1	21.4	23.0	23.6	23.7	22.9	21.0	16.6	13.1		
		MCDB	17.0	19.5	22.1	24.8	26.7	27.3	27.2	27.6	27.0	25.2	21.7	18.0		
	10%	WB	10.8	13.2	15.5	18.2	20.7	22.5	23.1	23.2	22.3	20.2	15.9	12.2		
		MCDB	15.8	18.5	21.7	23.5	25.8	26.5	26.5	26.8	26.0	23.7	20.6	16.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	14.3	14.2	14.0	11.4	8.7	7.0	7.7	8.2	11.1	13.0	14.1		
		MCDBR	16.1	16.1	15.7	16.4	13.5	10.0	8.3	8.8	9.6	12.3	14.0	15.6		
		MCWBR	8.7	7.6	6.6	6.4	5.8	4.1	3.8	3.7	4.2	5.5	6.7	8.2		
	5% WB	MCDBR	15.1	14.5	13.3	13.1	11.5	8.7	7.7	8.1	8.9	10.7	13.3	14.2		
		MCWBR	8.9	7.5	6.3	6.2	5.6	3.9	3.7	3.7	4.2	5.3	7.0	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.370	0.408	0.458	0.554	0.580	0.575	0.485	0.448	0.446	0.446	0.455	0.394		
	taud		2.205	2.118	1.991	1.796	1.799	1.867	2.174	2.297	2.257	2.133	1.986	2.137		
	Ebn at Noon		860	855	833	765	745	744	813	843	836	807	755	810		
	Edh at Noon		123	146	176	220	220	204	150	132	133	142	153	127		
(44) All-Sky Solar Radiation	RadAvg		3.81	4.56	5.60	6.17	6.01	5.29	4.38	4.49	4.44	4.77	4.27	3.80		
	RadStd		0.31	0.38	0.27	0.48	0.48	0.36	0.31	0.25	0.29	0.29	0.25	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page